

MECHATRONICS AND IOT

DESIGN AND DEVELOP A SMART INVENTORY MANAGEMENT SYSTEM FOR CONSTRUCTION SITES

ASSIGNMENT - 6

TEAM MEMBERS

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1. PROJECT OVERVIEW

The Smart Inventory Management System for Construction Sites is designed to efficiently monitor, track, and manage construction materials and inventory movement. Construction sites handle large quantities of materials such as cement, steel, bricks, pipes, tools, and other equipment. Manual inventory management can lead to material shortages, excess stock, loss, wastage, and inaccurate records.

The proposed system uses sensors, RFID, barcode/QR code technology, or a microcontroller-based system to identify and track materials. Whenever materials are received, stored, issued, or moved from one location to another, the inventory information is updated. The system keeps track of the available quantity and provides the current stock status.

The system can also provide alerts when the quantity of a material falls below the required level. This helps site managers arrange materials on time and avoid delays in construction activities. By reducing manual work and improving inventory visibility, the system helps achieve better resource utilization, reduces material wastage, and improves overall construction site management efficiency.

2. PROBLEM STATEMENT

Construction sites require large quantities of materials such as cement, steel, bricks, pipes, tools, and other resources. Managing these materials manually can result in inaccurate records, material loss, unnecessary wastage, overstocking, and unexpected shortages. Tracking the movement of materials between storage areas and work locations is also difficult using traditional methods. These problems can increase project costs and cause delays in construction activities.

Therefore, there is a need for a smart and efficient inventory management system that can continuously track material availability and movement at construction sites. The proposed system aims to monitor inventory levels, record material usage, identify stock shortages, and provide timely alerts when materials need to be replenished. This improves resource utilization, reduces wastage, minimizes manual record keeping, and supports efficient construction site management.

3. PROBLEM SOLUTION

The proposed solution is to develop a Smart Inventory Management System for construction sites to provide accurate and real-time monitoring of construction materials. The system is designed to overcome

the problems associated with traditional manual inventory management, such as incorrect stock records, material loss, excessive usage, wastage, and unexpected material shortages.

The system uses identification and sensing technologies such as RFID tags, barcode or QR codes, sensors, and a microcontroller to identify and track construction materials. Each material is assigned a unique identification so that its quantity and movement can be recorded. When materials enter the construction site, the system registers the received quantity and updates the inventory database.

Whenever a material is issued for construction work, transferred between storage locations, or returned to the storage area, the inventory information is updated. The system continuously calculates the available stock based on material entries and usage. This provides the site manager with accurate information about the current inventory.

A minimum stock level is assigned to important construction materials. When the available quantity falls below this predefined level, the system generates a warning or notification to indicate that the material needs to be replenished. This helps prevent construction delays caused by material shortages.

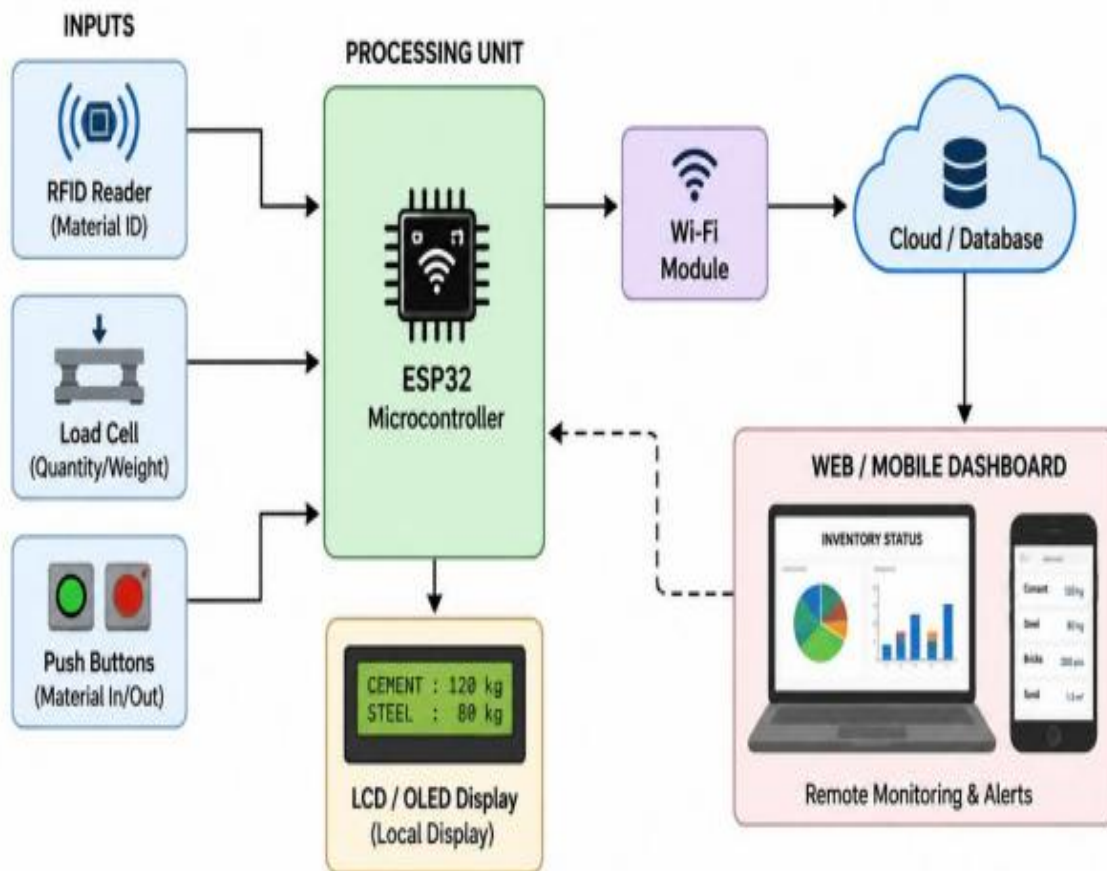
The system can also provide information about frequently used materials, material consumption, and inventory movement. An LCD display, computer, or mobile application can be used to view the inventory status. With IoT connectivity, the data can be stored on a cloud platform and monitored remotely by site managers.

The proposed system improves material utilization, reduces unnecessary wastage, minimizes manual record keeping, and provides better control over construction resources. It offers a cost-effective and scalable solution that can be implemented in small, medium, and large construction projects.

4. SYSTEM OF ARCHITECTURE

The Smart Inventory Management System consists of material identification, sensing, processing, storage, and monitoring units. RFID tags or barcode/QR codes are attached to construction materials for identification. When a material is received, issued, returned, or moved, the identification unit reads the material information and sends it to the microcontroller. The controller processes the inventory data and updates the available stock. The information can be stored in a database and displayed through an LCD, computer, or mobile application. If the stock level falls below the predefined minimum level, the system generates an alert to notify the site manager. IoT connectivity can also be added to enable real-time remote monitoring of construction materials.

SYSTEM ARCHITECTURE

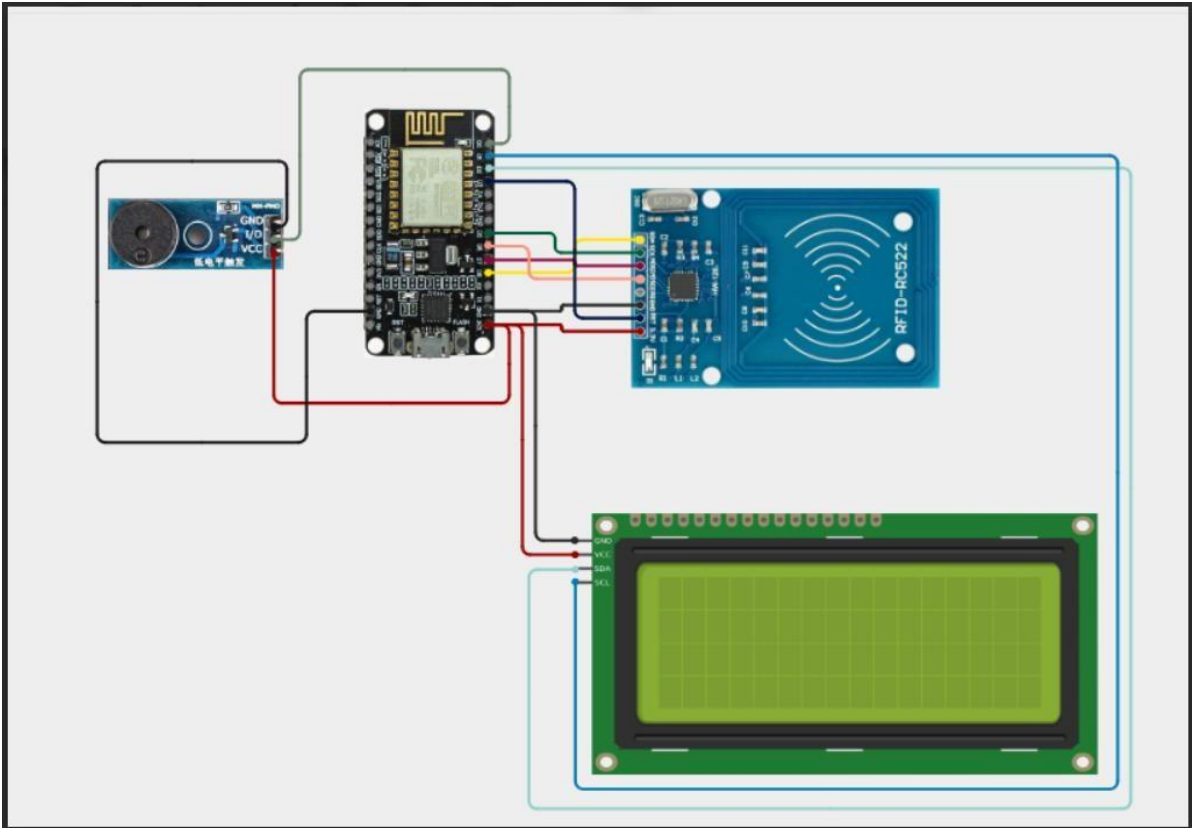


5. CIRCUIT TABLE

Component	Pin/Terminal	ESP8266 Connection	Purpose
RFID RC522	VCC	3.3V	Power Supply
LCD Display	VCC	3.3V	Safe indication
ESP8266	VIN/USB	5V USB	Hazardous indication
Buzzer	Positive (+)	GPIO 14	Alert

6.

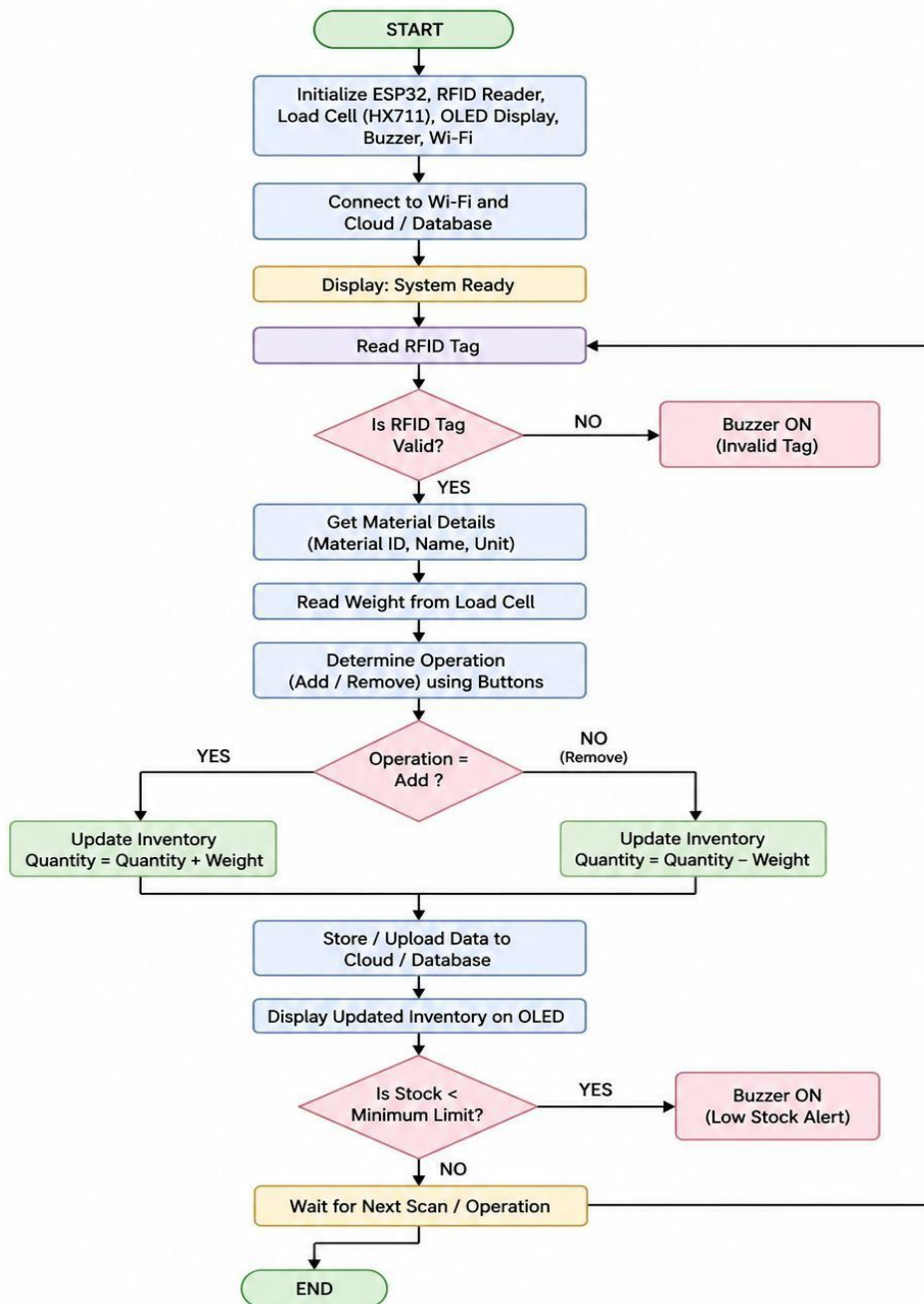
CIRCUIT DESIGN



The controller continuously checks the available quantity of each material and compares it with the predefined minimum stock level. If the available quantity is above the minimum level, the system displays the current stock status and continues monitoring. If the quantity falls below the minimum level, the system generates a low-stock alert to notify the site manager for timely replenishment. All material movement and inventory changes are recorded for future reference. The process is repeated continuously to provide accurate and real-time inventory management at the construction site.

7. ALGORITHM

ALGORITHM – SMART INVENTORY MANAGEMENT SYSTEM



8. CODING

```
#include <SPI.h>

#include <MFRC522.h>

#include <Wire.h>

#include <LiquidCrystal_I2C.h>

#include <ESP8266WiFi.h>

#include <ThingSpeak.h>

// =====

// WIFI SETTINGS

// =====

const char* ssid = "Xiaomi 11i";
const char* password = "12345678";
WiFiClient client;

// =====

// THINGSPEAK SETTINGS

// =====

unsigned long channelID = 3458770;
const char* writeAPIKey = "PH9Z3KRMKRAGSKBZ";

// =====

// RC522 RFID - ESP8266 GPIO PINS

// =====

#define SS_PIN 15    // D8
#define RST_PIN 0    // D3

MFRC522 mfrc522(SS_PIN, RST_PIN)

// =====
```

```
LiquidCrystal_I2C lcd(0x27, 16, 2);
```

```
// =====
```

// BUZZER - LOW ACTIVE

```
// =====
```

```
#define BUZZER_PIN 16 // D0
```

```
// LOW = Buzzer ON
```

```
// HIGH = Buzzer OFF
```

```
// =====
```

// SETUP

```
// =====
```

```
void setup() {
```

```
    //-----
```

// SERIAL MONITOR

```
    //  
    -----
```

```
    Serial.begin(9600);
```

```
    delay(1000);
```

```
    Serial.println();
```

```
    Serial.println("=====");
```

```
    Serial.println("    RFID SYSTEM");
```

```
    Serial.println("=====");
```

```
    // =====
```

// BUZZER SETUP

```
// =====
```

```
pinMode(BUZZER_PIN, OUTPUT);
```

// LOW ACTIVE BUZZER

```
// HIGH = OFF
```

```
digitalWrite(BUZZER_PIN, HIGH);
```

```
// =====  
  
Wire.begin(SDA_PIN, SCL_PIN);  
  
lcd.init();  
  
lcd.backlight();  
  
lcd.clear();  
  
lcd.setCursor(0, 0);  
  
lcd.print("RFID SYSTEM");  
  
lcd.setCursor(0, 1);  
  
lcd.print("Starting...");  
  
// =====
```

// CONNECT TO WIFI

```
// =====  
  
Serial.println();  
  
Serial.println("Connecting to WiFi...");  
  
lcd.clear();  
  
lcd.setCursor(0, 0);  
  
lcd.print("Connecting WiFi");  
  
WiFi.begin(ssid, password);  
  
while (WiFi.status() != WL_CONNECTED) {  
    delay(500);  
    Serial.print(".");  
}  
  
Serial.println();  
  
Serial.println("WiFi Connected");  
  
Serial.print("IP Address: ");  
  
Serial.println(WiFi.localIP());  
  
lcd.clear();  
  
lcd.setCursor(0, 0);  
  
lcd.print("WiFi Connected");
```

```
delay(1000);
```

```
// =====
```

// **START THINGSPEAK**

```
// =====
```

```
ThingSpeak.begin(client);
```

```
Serial.println("ThingSpeak Ready");
```

```
// =====
```

// **START RFID**

```
// =====
```

```
SPI.begin();
```

```
mfr522.PCD_Init();
```

```
delay(1000);
```

```
// =====
```

// **READY**

```
// =====
```

```
lcd.clear();
```

```
lcd.setCursor(0, 0);
```

```
lcd.print("RFID SYSTEM");
```

```
lcd.setCursor(0, 1);
```

```
lcd.print("Scan Your Card");
```

```
Serial.println("LCD Ready");
```

```
Serial.println("RC522 Ready");
```

```
Serial.println("ThingSpeak Ready");
```

```
Serial.println("Buzzer Ready");
```

```
Serial.println("Scan Your Card");
```

```
Serial.println("=====");
```

```
}
```

```
// =====
```

```
// =====  
  
void loop() {  
  
  //-----
```

// CHECK FOR NEW RFID CARD

```
// -----  
if (!mfrc522.PICC_IsNewCardPresent()) {  
  
  return;  
  
}  
  
// -----
```

// READ RFID CARD

```
// -----  
if (!mfrc522.PICC_ReadCardSerial()) {  
  
  return;  
  
}  
  
// -----
```

// CREATE UID STRING

```
// -----  
String uid = "";  
  
Serial.print("UID: ");  
  
for (byte i = 0; i < mfrc522.uid.size; i++) {  
  
  if (mfrc522.uid.uidByte[i] < 0x10) {  
  
    Serial.print("0");  
  
    uid += "0";  
  
  }  
  
  Serial.print(  
  
    mfrc522.uid.uidByte[i],  
  
    HEX  
  
  );  
  
  uid += String(  
  

```

```
        mfr522.uid.uidByte[i],  
        HEX  
    );  
}  
uid.toUpperCase();  
Serial.println();  
// =====
```

// DOOR

```
// =====  
if (uid == "BA8A8132") {  
    showKnownTag(  
        "Door",  
        uid,  
        1  
    );  
}  
// =====
```

// LAB

```
// =====  
else if (uid == "7BABD13E") {  
    showKnownTag(  
        "Lab",  
        uid,  
        2  
    );  
}  
// =====
```

// SATHEESH

```
// =====
```

```
else if (uid == "D61A9269") {
```

```
    showKnownTag(
```

```
        "Satheesh",
```

```
        uid,
```

```
        3
```

```
    );
```

```
}
```

```
// =====
```

// **SABEER**

```
// =====
```

```
else if (uid == "6F22CB48") {
```

```
    showKnownTag(
```

```
        "Sabeer",
```

```
        uid,
```

```
        4
```

```
    );
```

```
}
```

```
// =====
```

// **PRASANA**

```
// =====
```

```
else if (uid == "1F22EB48") {
```

```
    showKnownTag(
```

```
        "Prasana",
```

```
        uid,
```

```
        5
```

```
    );
```

```
}
```

```
// =====
```

// UNKNOWN CARD

```
// =====  
else {  
    showUnknownTag(uid);  
}  
  
// -----
```

// STOP RFID COMMUNICATION

```
// -----  
mfrc522.PICC_HaltA();  
mfrc522.PCD_StopCrypto1();  
  
//-----
```

// READY FOR NEXT CARD

```
// -----  
delay(1000);  
lcd.clear();  
lcd.setCursor(0, 0);  
lcd.print("RFID SYSTEM");  
lcd.setCursor(0, 1);  
lcd.print("Scan Your Card");  
}  
  
// =====
```

// FUNCTION: KNOWN RFID TAG

```
// =====  
  
void showKnownTag(  
    String name,  
    String uid,  
    int personID  
) {
```

```
// BUZZER OFF

// LOW ACTIVE BUZZER

// HIGH = OFF

// -----
digitalWrite(
  BUZZER_PIN,
  HIGH

);

// -----
```

// SERIAL OUTPUT

```
// -----
Serial.println();
Serial.print("Known RFID Tag: ");
Serial.println(name);
Serial.print("UID: ");
Serial.println(uid);
Serial.println("Access Granted");
Serial.println("-----");
//-----
```

// LCD OUTPUT

```
// -----
lcd.clear();
lcd.setCursor(0, 0);
lcd.print("Access Granted");
lcd.setCursor(0, 1);
lcd.print(name);
//-----
```

// SEND DATA TO THINGSPEAK

```
sendToThingSpeak(
```

```
    uid,
```

```
    personID,
```

```
    1
```

```
);
```

```
// -----
```

// KEEP RESULT ON LCD

```
// -----
```

```
delay(3000);
```

```
//-----
```

// MAKE SURE BUZZER IS OFF

```
// -----
```

```
digitalWrite(
```

```
    BUZZER_PIN,
```

```
    HIGH
```

```
);
```

```
}
```

```
// =====
```

// FUNCTION: UNKNOWN RFID TAG

```
// =====
```

```
void showUnknownTag(
```

```
    String uid
```

```
) {
```

```
// -----
```

// SERIAL OUTPUT

```
// -----
```

```
Serial.println();
```

```
Serial.print("Unknown RFID Tag:
```

```
Serial.println("Access Denied");

Serial.println("-----");

//-----
```

// LCD OUTPUT

```
//
// -----
lcd.clear();

lcd.setCursor(0, 0);

lcd.print("Unknown Tag");

lcd.setCursor(0, 1);

lcd.print("Access Denied");

//-----
```

// SEND DATA TO THINGSPEAK

```
//
// -----
sendToThingSpeak(

uid,

0,

0

);

// -----
```

// BUZZER - LOW ACTIVE

```
// 3 BEEPS

// -----

for (int i = 0; i < 3; i++) {

// BUZZER ON

digitalWrite(

BUZZER_PIN,

LOW

);
```

// BUZZER OFF

```
digitalWrite(  
    BUZZER_PIN,  
    HIGH  
);  
  
delay(300);  
  
}  
  
// -----
```

// MAKE SURE BUZZER IS OFF

```
// -----  
digitalWrite(  
    BUZZER_PIN,  
    HIGH  
  
);  
  
// -----
```

// KEEP RESULT ON LCD

```
// -----  
delay(2000);  
  
}  
  
// =====
```

// FUNCTION: SEND DATA TO THINGSPEAK

```
// =====  
  
void sendToThingSpeak(  
    String uid,  
    int personID,  
    int accessStatus  
) {
```

```
//-----  
if (WiFi.status() != WL_CONNECTED) {  
    Serial.println(  
        "WiFi Disconnected"  
    );  
    return;  
}  
  
//-----  
// FIELD 1 = RFID UID  
  
//-----  
ThingSpeak.setField(  
    1,  
    uid  
);  
  
//-----  
// FIELD 2 = PERSON ID  
  
//-----  
ThingSpeak.setField(  
    2,  
    personID  
);  
  
//-----  
// FIELD 3 = ACCESS STATUS  
  
//-----  
ThingSpeak.setField(  
    3,  
    accessStatus  
);  
  
//-----
```

```
// WRITE DATA TO THINGSPEAK
```

```
//
```

```
-----  
int response =
```

```
    ThingSpeak.writeFields(  
        channelID,
```

```
        writeAPIKey
```

```
    );
```

```
//
```

```
// CHECK RESPONSE
```

```
//
```

```
-----  
if (response ==
```

```
    200) {
```

```
    Serial.println(  
        "ThingSpeak Update Successful"
```

```
    );
```

```
};
```

```
else
```

```
{
```

```
    Se
```

```
    ria
```

```
    l.p
```

```
    rin
```

```
    t(  
        "ThingSpeak Error: "
```

```
    );
```

```
    Serial.println(response);
```

```
} SYSTEM WORKING
```

9. SYSTEM WORKING

The Smart Inventory Management System works by continuously tracking the quantity and movement of construction materials at the site. Each material is assigned a unique RFID tag, barcode, or QR code, which allows the system to identify the material and record its details. When materials are received at the construction site, the identification code is scanned and the material name, quantity, and storage location are entered into the system.

When a material is taken from the storage area for construction work, the code is scanned again. The system automatically reduces the used quantity from the available stock. Similarly, when materials are returned or transferred to another storage location, the inventory record is updated accordingly.

The microcontroller processes all the inventory information and continuously checks the available stock. A predefined minimum stock level is maintained for each material. If the quantity falls below this level, the system generates a low-stock alert to inform the site manager that the material needs to be replenished.

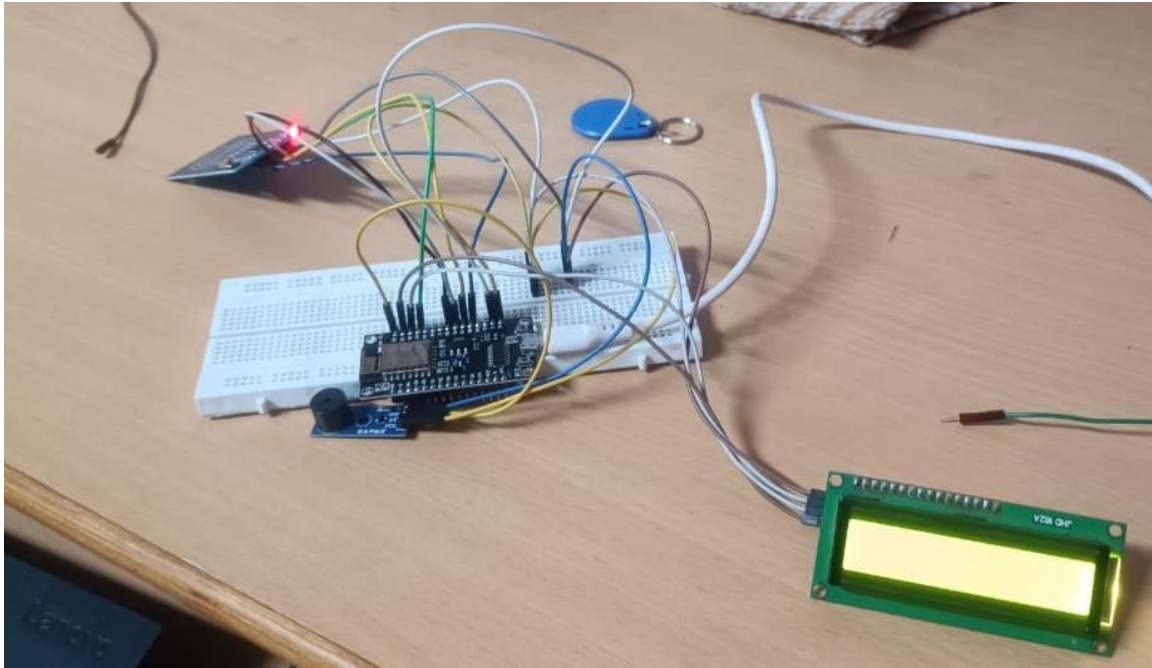
The current inventory status can be displayed on an LCD, computer, or mobile application. With IoT connectivity, inventory data can also be transmitted to a cloud platform for remote monitoring. This continuous monitoring helps reduce material wastage, prevent shortages, improve inventory accuracy, and ensure efficient utilization of construction resources.

10. BILL OF MATERIAL

S. no	Component	Specification	Quantity	Cost
1	ESP8266	Node MCU ESP8266	1	Rs. 450
2	RFID Reader	RC522 RFID Module	1	Rs. 150
3	OLED Display	0.96" I ² C, SSD1306, 128×64	1	Rs. 150
5	Breadboard	Standard 830-point	1	Rs. 100
6	Jumper Wire	Male-Male/Male-Female	1 set	Rs. 50
			Total	Rs. 900

11. PROTOTYPE IMPLEMENTATION

- The prototype was successfully implemented using an ESP32 microcontroller, RFID RC522 reader, load cell with HX711 module, OLED display, push buttons, and buzzer. The ESP32 acts as the main controller and manages the identification, measurement, and inventory tracking operations.
- The RFID RC522 reader is used to identify construction materials through unique RFID tags. Each material is assigned a specific RFID tag, allowing the system to recognize the material whenever it is added, removed, or checked at the construction site.
- The load cell is connected to the HX711 amplifier module and is used to measure the weight of the material. The measured weight is processed by the ESP32 and used to determine the quantity of material available in the inventory.
- The ADD and REMOVE push buttons allow the operator to select the required inventory operation. When the ADD button is pressed, the system records the material being added and updates the inventory quantity. When the REMOVE button is selected, the system records the material being removed and decreases the available quantity.
- The ESP32 processes the RFID identification and load-cell readings and updates the inventory information accordingly. The current material name, weight, and inventory status are displayed on the OLED display, allowing the site operator to easily monitor the available resources.
- A buzzer is included in the prototype to provide an audible warning. It is activated when an invalid RFID tag is detected or when the inventory level falls below the predefined minimum stock level.
- The system can maintain separate inventory information for different construction materials and update the records whenever material movement occurs. This helps reduce manual record keeping and improves inventory accuracy.
- The ESP32's built-in Wi-Fi capability can be used to transmit inventory information to a cloud database or IoT platform. This allows authorized users to monitor material availability and movement remotely.
- The prototype provides a practical method for tracking construction materials, monitoring stock levels, reducing material wastage, and preventing unexpected shortages at construction sites.



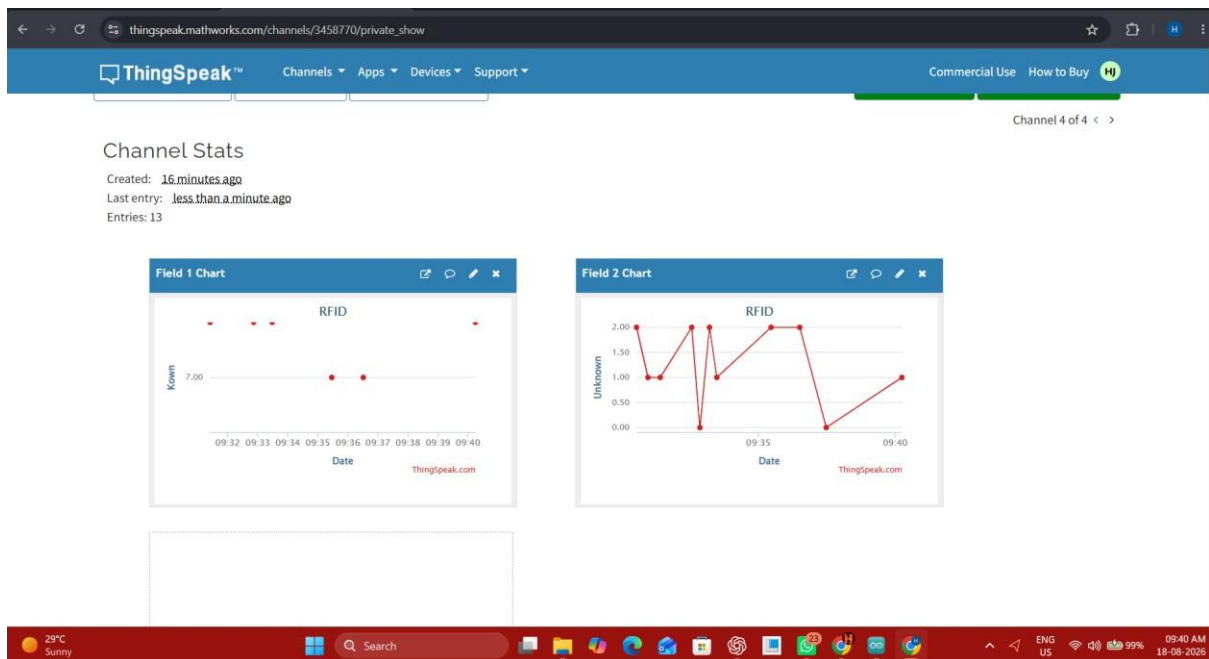
12. RESULT AND PERFORMANCE

The Smart Inventory Management System for Construction Sites was successfully implemented and tested using an ESP32 microcontroller, RFID RC522 reader, load cell with HX711 module, OLED display, push buttons, and buzzer. The prototype successfully identified construction materials, measured material weight, and updated inventory information based on the selected ADD or REMOVE operation.

The RFID system provided quick and reliable identification of materials using unique RFID tags. The load cell successfully measured changes in material weight when materials were added or removed. The ESP32 processed the RFID and weight data and updated the inventory status in real time.

The system provided:

- Real-time monitoring of construction material inventory.
- Automatic identification of materials using RFID tags.
- Measurement of material quantity using a load cell and HX711 module.
- Easy ADD and REMOVE operations using push buttons.
- Real-time display of material information on the OLED display.
- Low-stock alerts when the inventory falls below the predefined limit.
- Buzzer alerts for invalid RFID tags and low inventory conditions.
- Reduced manual inventory recording.
- Improved tracking of material movement and usage.
- Wi-Fi connectivity for possible cloud-based remote monitoring.
- Better control over material availability and utilization.



Overall, the prototype demonstrated satisfactory performance in tracking construction materials and updating inventory levels. The system can help reduce material wastage, prevent unexpected shortages, improve inventory accuracy, and increase resource utilization at construction sites. It can be further improved by adding a mobile application, cloud database, multiple RFID readers, and advanced inventory analytics.